

ELITE MATHEMATICS

Pearson Edexcel International A Level

# Pure 2

## Adaptive Classified

Chapter 08 · Modelling with Sequences and Series

### With Answers

Questions with full worked solutions

- 27 pages
- Past papers up to January 2026

PURE 2 | TOPIC 08

# Modelling with Sequences & Series

Your classified route

**10**

QUESTIONS

**83**

MARKS

**4**

FAMILIES

**5**

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

## TOPIC 08 | FAMILY 01

# Model Fitting

**WHAT THIS FAMILY ASKS**

Recognise the model fitting demand and complete the required reasoning in a clear, exam-ready order.

**VARIANT ROUTE**

**V01** core family move

**2 classified questions**

17 total marks

T08

08\_FIT

08\_FIT\_CORE

CHAPTER

**Modelling with Sequences & Series**

FAMILY

**Model fitting**

VARIANT

**core family move****9 marks**

WMA12/01 MAY/JUNE 2022

**Question 9**

Past-paper question

WMA12-01\_2022\_MayJune\_Q09

9. A scientist is using carbon-14 dating to determine the age of some wooden items.

The equation for carbon-14 dating an item is given by

$$N = k\lambda^t$$

where

- $N$  grams is the amount of carbon-14 **currently** present in the item
- $k$  grams was the **initial** amount of carbon-14 present in the item
- $t$  is the number of years since the item was made
- $\lambda$  is a constant, with  $0 < \lambda < 1$

- (a) Sketch the graph of  $N$  against  $t$  for  $k = 1$

**(2)**

Given that it takes 5700 years for the amount of carbon-14 to reduce to half its initial value,

- (b) show that the value of the constant  $\lambda$  is 0.999878 to 6 decimal places.

**(2)**

Given that Item  $A$

- is known to have had 15 grams of carbon-14 present initially
- is thought to be 3250 years old

- (c) calculate, to 3 significant figures, how much carbon-14 the equation predicts is currently in Item  $A$ .

**(2)**

Item  $B$  is known to have initially had 25 grams of carbon-14 present, but only 18 grams now remain.

- (d) Use algebra to calculate the age of Item  $B$  to the nearest 100 years.

**(3)**

T08 Modelling with  
Sequences & Series

08\_FIT Model fitting

08\_FIT\_CORE core family  
move

## WORKED SOLUTION

## Question 9

WMA12/01 May/June 2022

WMA12-01\_2022\_MayJune\_Q09

1

**Sketch the model for k equals one**

For  $k = 1$ ,  $N = \lambda^t$  with  $0 < \lambda < 1$ . The graph starts at  $(0, 1)$ , decreases as  $t$  increases, and tends towards 0.

2

**Use the half-life information**

After 5700 years,  $N = \frac{k}{2}$ , so  $k\lambda^{5700} = \frac{k}{2}$ .

3

**Find lambda**

$\lambda^{5700} = \frac{1}{2}$ , hence  $\lambda = \left(\frac{1}{2}\right)^{1/5700} = 0.999878 \dots$ , so  $\lambda = 0.999878$  to 6 decimal places.

4

**Model item A**

For item A,  $k = 15$  and  $t = 3250$ , so  $N = 15\lambda^{3250}$ .

5

**Calculate item A**

$N = 15(0.9998784026 \dots)^{3250} = 10.103 \dots$ , so the current amount is 10.1 g to 3 significant figures.

6

**Set up item B**

For item B,  $18 = 25\lambda^t$ , so  $\lambda^t = \frac{18}{25}$ .

7

**Take logarithms**

$t \log \lambda = \log \left(\frac{18}{25}\right)$ , so  $t = \frac{\log(18/25)}{\log \lambda}$ .

8

**Round the age**

Using  $\lambda = (1/2)^{1/5700}$  gives  $t = 2701 \dots$ , so the age is 2700 years to the nearest 100 years.

**FINAL ANSWER**

WMA12-01\_2022\_MayJune\_Q09

(a) Decreasing exponential through  $(0, 1)$ . (b)  $\lambda = 0.999878$ .  
(c) 10.1 g. (d) 2700 years.

T08

08\_FIT

08\_FIT\_CORE

CHAPTER

**Modelling with Sequences & Series**

FAMILY

**Model fitting**

VARIANT

**core family move****8 marks**

WMA12/01 MAY/JUNE 2024

**Question 10**

Past-paper question

WMA12-01\_2024\_MayJune\_Q10

10. **In this question you must show all stages of your working.  
Solutions relying entirely on calculator technology are not acceptable.**

The number of dormice and the number of voles on an island are being monitored.

Initially there are 2000 dormice on the island.

A model predicts that the number of dormice will increase by 3% each year, so that the numbers of dormice on the island at the end of each year form a geometric sequence.

- (a) Find, according to the model, the number of dormice on the island 6 years after monitoring began. Give your answer to 3 significant figures.

**(2)**

The number of voles on the island is being monitored over the same period of time.

Given that

- 4 years after monitoring began there were 3690 voles on the island
- 7 years after monitoring began there were 3470 voles on the island
- the number of voles on the island at the end of each year is modelled as a geometric sequence

- (b) find the equation of this model in the form

$$N = ab^t$$

where  $N$  is the number of voles,  $t$  years after monitoring began and  $a$  and  $b$  are constants. Give the value of  $a$  and the value of  $b$  to 2 significant figures.

**(3)**

When  $t = T$ , the number of dormice on the island is equal to the number of voles on the island.

- (c) Find, according to the models, the value of  $T$ , giving your answer to one decimal place.

**(3)**

T08 Modelling with  
Sequences & Series

08\_FIT Model fitting

08\_FIT\_CORE core family  
move

## WORKED SOLUTION

# Question 10

WMA12/01 May/June 2024

WMA12-01\_2024\_MayJune\_Q10

1

### Model the dormice

The dormice model is  $D = 2000(1.03)^t$ .

2

### Find the dormice after six years

$D = 2000(1.03)^6 = 2388.10 \dots$ , so there are 2390 dormice to 3 significant figures.

3

### Set up the vole model

Let the vole model be  $N = ab^t$ .

4

### Use the two data points

At  $t = 4$ ,  $ab^4 = 3690$ . At  $t = 7$ ,  $ab^7 = 3470$ .

5

### Find b

Dividing gives  $b^3 = \frac{3470}{3690}$ , so  $b = 0.9797 \dots$ , which is 0.98 to 2 significant figures.

6

### Find a

$a = \frac{3690}{b^4} = 4005.18 \dots$ , which is 4000 to 2 significant figures.

7

### Write the vole model

So the model is approximately  $N = 4000(0.98)^t$ .

8

### Set the two models equal

Use  $2000(1.03)^T = ab^T$  with the unrounded constants from the vole model.

9

### Solve using logarithms

$\left(\frac{1.03}{b}\right)^T = \frac{a}{2000}$ , so  $T = \frac{\log(a/2000)}{\log(1.03/b)}$ .

10

### Calculate T

Substituting  $a = 4005.18 \dots$  and  $b = 0.9797 \dots$  gives  $T = 13.875 \dots$ , so  $T = 13.9$  to one decimal place.

**FINAL ANSWER**

WMA12-01\_2024\_MayJune\_Q10

(a) 2390 **dormice.**      (b)  $N = 4000(0.98)^t$  **to 2 s.f. constants.**      (c)  $T = 13.9.$

DR ESLAM AHMED

## TOPIC 08 | FAMILY 02

# Interpretation

**WHAT THIS FAMILY ASKS**

Recognise the interpretation demand and complete the required reasoning in a clear, exam-ready order.

**VARIANT ROUTE**

**V01** For 'at least' or 'exceeds', test the boundary integer.

**V02** If the model starts at  $n=0$ , translate  $n$  into calendar time.

**3 classified questions**

24 total marks

T08

08\_INTERP

08\_INTERP\_HV01

CHAPTER

**Modelling with Sequences & Series**

FAMILY

**Interpretation**

VARIANT

**For 'at least' or 'exceeds', test the boundary integer.****8 marks**

WMA12/01 JANUARY 2020

**Question 5**

Past-paper question

WMA12-01\_2020\_Jan\_Q05

5. In this question you must show all stages of your working.

**Solutions relying on calculator technology are not acceptable.**

A colony of bees is being studied.

The number of bees in the colony at the start of the study was 30 000

Three years after the start of the study, the number of bees in the colony is 34 000

A model predicts that the number of bees in the colony will increase by  $p\%$  each year, so that the number of bees in the colony at the end of each year of study forms a geometric sequence.

Assuming the model,

(a) find the value of  $p$ , giving your answer to 2 decimal places.

**(3)**

According to the model, at the end of  $N$  years of study the number of bees in the colony exceeds 75 000

(b) Find, showing all steps in your working, the smallest integer value of  $N$ .

**(5)**

T08 Modelling with  
Sequences & Series

08\_INTERP Interpretation

08\_INTERP\_HV01 For 'at  
least' or 'exceeds', test the  
boundary integer.

## WORKED SOLUTION

### Question 5

WMA12/01 January 2020

WMA12-01\_2020\_Jan\_Q05

1

#### Set up the growth ratio

Let the yearly ratio be  $r$ . After 3 years,  $30000r^3 = 34000$ .

2

#### Find $p$

$r^3 = \frac{17}{15}$ , so  $r = \left(\frac{17}{15}\right)^{1/3} \approx 1.04257$ . Therefore  $p = (r - 1) \times 100 \approx 4.26$ .

3

#### Set up the inequality

After  $N$  years the model gives  $30000r^N$ . We need  $30000r^N > 75000$ , so  $r^N > 2.5$ .

4

#### Take logarithms

$N \log r > \log(2.5)$ , so  $N > \frac{\log(2.5)}{\log r}$ .

5

#### Find the smallest integer

Using  $r \approx 1.04257$ ,  $\frac{\log(2.5)}{\log r} \approx 21.98$ . Hence the smallest integer value is  $N = 22$ .

#### FINAL ANSWER

(a)  $p = 4.26$ . (b)  $N = 22$ .

WMA12-01\_2020\_Jan\_Q05

T08

08\_INTERP

08\_INTERP\_HV01

## CHAPTER

## Modelling with Sequences &amp; Series

## FAMILY

## Interpretation

## VARIANT

For 'at least' or 'exceeds', test the boundary integer.

9 marks

WMA12/01 JANUARY 2022

## Question 8

Past-paper question

WMA12-01\_2022\_Jan\_Q08

8. A metal post is repeatedly hit in order to drive it into the ground.

Given that

- on the 1st hit, the post is driven 100 mm into the ground
- on the 2nd hit, the post is driven an **additional** 98 mm into the ground
- on the 3rd hit, the post is driven an **additional** 96 mm into the ground
- the **additional** distances the post travels on each subsequent hit form an arithmetic sequence

- (a) show that the post is driven an **additional** 62 mm into the ground with the 20th hit.

(1)

- (b) Find the **total distance** that the post has been driven into the ground after 20 hits.

(2)

Given that for each subsequent hit after the 20th hit

- the **additional** distances the post travels form a geometric sequence with common ratio  $r$
- on the 22nd hit, the post is driven an **additional** 60 mm into the ground

- (c) find the value of  $r$ , giving your answer to 3 decimal places.

(2)

After a total of  $N$  hits, the post will have been driven more than 3 m into the ground.

- (d) Find, showing all steps in your working, the smallest possible value of  $N$ .

(4)

T08 Modelling with  
Sequences & Series

08\_INTERP Interpretation

08\_INTERP\_HV01 For 'at  
least' or 'exceeds', test the  
boundary integer.

## WORKED SOLUTION

### Question 8

WMA12/01 January 2022

WMA12-01\_2022\_Jan\_Q08

1

#### Use the arithmetic rule

For the first 20 hits, the additional distances form an arithmetic sequence with first term 100 and common difference  $-2$ .

2

#### Show the twentieth hit

The twentieth additional distance is  $100 + 19(-2) = 62$  mm.

3

#### Sum the first twenty hits

$$S_{20} = \frac{20}{2}(100 + 62) = 1620 \text{ mm.}$$

4

#### Use the geometric information

After the twentieth hit, the distances follow the same ratio, so the twenty second distance is  $62r^2$ .

5

#### Find $r$

$$62r^2 = 60, \text{ hence } r = \sqrt{\frac{30}{31}} = 0.984 \text{ to 3 decimal places.}$$

6

#### Set up the remaining distance

More than 3 m means more than 3000 mm. After 20 hits, the remaining amount to exceed is  $3000 - 1620 = 1380$  mm.

7

#### Sum the geometric tail

$$\text{If there are } m \text{ further hits, their total is } 62r + 62r^2 + \dots + 62r^m = 62r \frac{1-r^{m+1}}{1-r}.$$

8

#### Find the first $m$ that works

Using  $r = \sqrt{30/31}$ ,  $62r \frac{1-r^{27}}{1-r}$  is not enough, but  $62r \frac{1-r^{28}}{1-r}$  makes the total exceed 3000 mm.

9

#### State $N$

Therefore  $m = 28$  further hits are needed after the first 20, so  $N = 20 + 28 = 48$ .

**FINAL ANSWER**

WMA12-01\_2022\_Jan\_Q08

(a) **Proven:** 62 mm.    (b) 1620 mm.    (c)  $r = 0.984$ .    (d)  $N = 48$ .

DR ESLAM AHMED

T08

08\_INTERP

08\_INTERP\_HV02

## CHAPTER

**Modelling with Sequences & Series**

## FAMILY

**Interpretation**

## VARIANT

If the model starts at  $t=0$ ,  
translate  $t(n)$  into calendar  
time.

**7 marks**

WMA12/01 MAY/JUNE 2021

**Question 1**

Past-paper question

WMA12-01\_2021\_MayJune\_Q01

1. Adina is saving money to buy a new computer. She saves £5 in week 1, £5.25 in week 2, £5.50 in week 3 and so on until she has enough money, in total, to buy the computer.

She decides to model her savings using either an arithmetic series or a geometric series.

Using the information given,

- (a) (i) state with a reason whether an arithmetic series or a geometric series should be used,
- (ii) write down an expression, in terms of  $n$ , for the amount, in pounds (£), saved in week  $n$ .

**(3)**

Given that the computer Adina wants to buy costs £350

- (b) find the number of weeks it will take for Adina to save enough money to buy the computer.

**(4)**

T08 Modelling with  
Sequences & Series

08\_INTERP Interpretation

08\_INTERP\_HV02 If the model  
starts at  $t=0$ , translate  $t(n)$   
into calendar time.

## WORKED SOLUTION

### Question 1

WMA12/01 May/June 2021

WMA12-01\_2021\_MayJune\_Q01

1

#### Choose the model

The weekly amounts are 5, 5.25, 5.50, and so on. The increase is a constant 0.25, so an arithmetic series should be used.

2

#### Write the $n$ th amount

The first amount is 5 and the common difference is 0.25, so the amount saved in week  $n$  is  $5 + 0.25(n - 1)$ .

3

#### Set up the total

After  $n$  weeks,  $S_n = \frac{n}{2}\{2(5) + (n - 1)(0.25)\}$ .

4

#### Form the inequality

Adina needs  $S_n \geq 350$ , so  $\frac{n}{2}(9.75 + 0.25n) \geq 350$ .

5

#### Solve for $n$

This is  $n^2 + 39n - 2800 \geq 0$ . The positive root is 36.89 ..., so the first whole number that works is  $n = 37$ .

6

#### Check the week

After 36 weeks the total is below 350, and after 37 weeks it is 351.50, so it takes 37 weeks.

#### FINAL ANSWER

WMA12-01\_2021\_MayJune\_Q01

(a)(i) **Arithmetic series.** (a)(ii)  $5 + 0.25(n - 1)$ . (b) **37 weeks.**

## TOPIC 08 | FAMILY 03

# Limiting Models

**WHAT THIS FAMILY ASKS**

Recognise the limiting models demand and complete the required reasoning in a clear, exam-ready order.

**VARIANT ROUTE**

**V01**    core family move

**2 classified questions**

21 total marks

T08

08\_LIMIT

08\_LIMIT\_CORE

CHAPTER

**Modelling with Sequences & Series**

FAMILY

**Limiting models**

VARIANT

**core family move****11 marks**

WMA12/01 JANUARY 2021

**Question 10**

Past-paper question

WMA12-01\_2021\_Jan\_Q10

**10. In this question you must show detailed reasoning.**

Owen wants to train for 12 weeks in preparation for running a marathon.

During the 12-week period he will run every Sunday and every Wednesday.

- On Sunday in week 1 he will run 15 km
- On Sunday in week 12 he will run 37 km

He considers two different 12-week training plans.

In training plan *A*, he will increase the distance he runs each Sunday by the same amount.

- (a) Calculate the distance he will run on Sunday in week 5 under training plan *A*.

**(3)**

In training plan *B*, he will increase the distance he runs each Sunday by the same percentage.

- (b) Calculate the distance he will run on Sunday in week 5 under training plan *B*.  
Give your answer in km to one decimal place.

**(3)**

Owen will also run a fixed distance,  $x$  km, each Wednesday over the 12-week period.

Given that

- $x$  is an integer
- the total distance that Owen will run on Sundays and Wednesdays over the 12 weeks will not exceed 360 km

- (c) (i) find the maximum value of  $x$ , if he uses training plan *A*,  
(ii) find the maximum value of  $x$ , if he uses training plan *B*.

**(5)****(Total 11 marks)**

T08 Modelling with  
Sequences & Series

08\_LIMIT Limiting models

08\_LIMIT\_CORE core family  
move

## WORKED SOLUTION

## Question 10

WMA12/01 January 2021

WMA12-01\_2021\_Jan\_Q10

1

**Find the arithmetic difference**

Plan A has 12 Sunday runs from 15 km to 37 km, so there are 11 equal increases. The common difference is  $\frac{37-15}{11} = 2$ .

2

**Find week five for plan A**

Week 5 is the fifth term, so the distance is  $15 + 4(2) = 23$  km.

3

**Find the geometric ratio**

For plan B,  $15r^{11} = 37$ , so  $r = \left(\frac{37}{15}\right)^{1/11}$ .

4

**Find week five for plan B**

The fifth Sunday distance is  $15r^4 = 15\left(\frac{37}{15}\right)^{4/11} = 20.829 \dots$ , so it is 20.8 km.

5

**Sum the Sunday runs for plan A**

The arithmetic total is  $\frac{12}{2}(15 + 37) = 312$  km.

6

**Find the Wednesday limit for plan A**

The total distance condition gives  $312 + 12x \leq 360$ , so  $x \leq 4$ . The maximum integer is 4.

7

**Sum the Sunday runs for plan B**

The geometric total is  $15\frac{r^{12}-1}{r-1} = 294.185 \dots$  km.

8

**Find the Wednesday limit for plan B**

Now  $294.185 \dots + 12x \leq 360$ , so  $x \leq 5.484 \dots$ . The maximum integer is 5.

**FINAL ANSWER**

WMA12-01\_2021\_Jan\_Q10

(a) 23 km. (b) 20.8 km. (c)(i)  $x = 4$ . (c)(ii)  $x = 5$ .

T08

08\_LIMIT

08\_LIMIT\_CORE

CHAPTER

**Modelling with Sequences & Series**

FAMILY

**Limiting models**

VARIANT

**core family move****10 marks****WMA12/01 JANUARY 2026****Question 9**

Past-paper question

WMA12-01\_2026\_Jan\_Q09

9. A company mines lithium.

- in year 1, the company mined 12 400 tonnes of lithium
- in year 15, the company is predicted to mine 7 500 tonnes of lithium

Model *A* assumes that the mass of lithium the company mines will decrease by the same amount each year.

According to model *A*,

(a) find the mass of lithium the company will mine in year 5 (3)

(b) calculate the **total** mass of lithium the company will mine from year 1 to year 15 inclusive. (2)

Model *B* assumes that the mass of lithium the company mines will decrease by the same percentage each year.

(c) Find, according to model *B*, the mass of lithium the company will mine in year 10. Give the answer to the nearest 10 tonnes. (3)

According to model *B*, there is a limit to the **total** mass of lithium that can be mined.

(d) Calculate the value of this limit. Give the answer to the nearest 100 tonnes. (2)

T08 Modelling with  
Sequences & Series

08\_LIMIT Limiting models

08\_LIMIT\_CORE core family  
move

## WORKED SOLUTION

## Question 9

WMA12/01 January 2026

WMA12-01\_2026\_Jan\_Q09

1

**Model A as an arithmetic sequence**

For Model A, let the common difference be  $d$ . Since year 15 is the 15th term,  $7500 = 12400 + 14d$ .

2

**Find year 5 for Model A**

$d = \frac{7500-12400}{14} = -350$ . Year 5 is  $12400 + 4(-350) = 11000$  tonnes.

3

**Find the total from year 1 to year 15**

$S_{15} = \frac{15}{2}(12400 + 7500) = 149250$  tonnes.

4

**Model B as a geometric sequence**

Let the common ratio be  $r$ . Then  $7500 = 12400r^{14}$ , so  $r = \left(\frac{7500}{12400}\right)^{1/14} \approx 0.964723$ .

5

**Find year 10 for Model B**

Year 10 is  $12400r^9 \approx 8975.26$ , which is 8980 tonnes to the nearest 10 tonnes.

6

**Find the limiting total**

Since  $0 < r < 1$ , the sum to infinity is  $S_{\infty} = \frac{12400}{1-r} \approx 351508.1$ , so the limit is 351500 tonnes to the nearest 100 tonnes.

**FINAL ANSWER**

WMA12-01\_2026\_Jan\_Q09

(a) 11000 tonnes. (b) 149250 tonnes. (c) 8980 tonnes. (d) 351500 tonnes.

## TOPIC 08 | FAMILY 04

# Model Setup

**WHAT THIS FAMILY ASKS**

Recognise the model setup demand and complete the required reasoning in a clear, exam-ready order.

**VARIANT ROUTE**

**V01** core family move

**3 classified questions**

21 total marks

T08

08\_MODEL

08\_MODEL\_CORE

CHAPTER

**Modelling with Sequences & Series**

FAMILY

**Model setup**

VARIANT

**core family move****6 marks****WMA12/01 OCTOBER 2019****Question 2**

Past-paper question

WMA12-01\_2019\_Oct\_Q02

2. The adult population of a town at the start of 2019 is 25 000

A model predicts that the adult population will increase by 2% each year, so that the number of adults in the population at the start of each year following 2019 will form a geometric sequence.

- (a) Find, according to the model, the adult population of the town at the start of 2032 **(3)**

It is also modelled that every member of the adult population gives £5 to local charity at the start of each year.

- (b) Find, according to these models, the total amount of money that would be given to local charity by the adult population of the town from 2019 to 2032 inclusive. Give your answer to the nearest £1 000 **(3)**

T08 Modelling with  
Sequences & Series

08\_MODEL Model setup

08\_MODEL\_CORE core  
family move

## WORKED SOLUTION

### Question 2

WMA12/01 October 2019

WMA12-01\_2019\_Oct\_Q02

1

#### Identify the common ratio

An increase of 2% each year gives the geometric ratio  $r = 1.02$ .

2

#### Count the years to 2032

From the start of 2019 to the start of 2032 there are 13 yearly increases, so the population is  $25000(1.02)^{13}$ .

3

#### Calculate the 2032 population

$25000(1.02)^{13} \approx 32340.7$ , so the model gives about 32341 adults.

4

#### Set up the total number of adult payments

From 2019 to 2032 inclusive there are 14 yearly populations, so the total number of adult payments is  $25000 \frac{1.02^{14}-1}{1.02-1}$ .

5

#### Multiply by the charity amount

The total money is  $5 \times 25000 \frac{1.02^{14}-1}{0.02} \approx 1\,996\,766$ , so this is about £1,996,766.

6

#### Round as requested

To the nearest £1000, this is £1,997,000.

#### FINAL ANSWER

(a) **32341 adults, approximately.** (b) **£1,997,000.**

WMA12-01\_2019\_Oct\_Q02

T08

08\_MODEL

08\_MODEL\_CORE

CHAPTER

**Modelling with Sequences & Series**

FAMILY

**Model setup**

VARIANT

**core family move****9 marks**

WMA12/01 JANUARY 2024

**Question 7**

Past-paper question

WMA12-01\_2024\_Jan\_Q07

7. Wheat is grown on a farm.

- In year 1, the farm produced 300 tonnes of wheat.
- In year 12, the farm is predicted to produce 4 000 tonnes of wheat.

Model  $A$  assumes that the amount of wheat produced on the farm will increase by the same amount each year.

- (a) Using model  $A$ , find the amount of wheat produced on the farm in year 4.  
Give your answer to the nearest 10 tonnes.

**(3)**

Model  $B$  assumes that the amount of wheat produced on the farm will increase by the same percentage each year.

- (b) Using model  $B$ , find the amount of wheat produced on the farm in year 2.  
Give your answer to the nearest 10 tonnes.

**(3)**

- (c) Calculate, according to the two models, the difference between the total amounts of wheat predicted to be produced on the farm from year 1 to year 12 inclusive.  
Give your answer to the nearest 10 tonnes.

**(3)**

T08 Modelling with  
Sequences & Series

08\_MODEL Model setup

08\_MODEL\_CORE core  
family move

## WORKED SOLUTION

## Question 7

WMA12/01 January 2024

WMA12-01\_2024\_Jan\_Q07

1

**Model A common difference**

Model A is arithmetic. From year 1 to year 12 there are 11 equal increases, so  $d = \frac{4000-300}{11} = \frac{3700}{11}$ .

2

**Find year 4 in model A**

Year 4 is  $300 + 3d = 300 + 3\left(\frac{3700}{11}\right) = 1309.09 \dots$ , so the amount is 1310 tonnes to the nearest 10 tonnes.

3

**Model B common ratio**

Model B is geometric. If the ratio is  $r$ , then  $300r^{11} = 4000$ .

4

**Find the ratio**

$$r = \left(\frac{4000}{300}\right)^{1/11} = 1.2655 \dots$$

5

**Find year 2 in model B**

Year 2 is  $300r = 379.65 \dots$ , so the amount is 380 tonnes to the nearest 10 tonnes.

6

**Total for model A**

$$S_A = \frac{12}{2}(300 + 4000) = 25800 \text{ tonnes.}$$

7

**Total for model B**

$$S_B = 300 \frac{r^{12}-1}{r-1} = 17935.20 \dots \text{ tonnes.}$$

8

**Find the difference**

The difference is  $25800 - 17935.20 \dots = 7864.79 \dots$ , which is 7860 tonnes to the nearest 10 tonnes.

**FINAL ANSWER**

(a) 1310 tonnes. (b) 380 tonnes. (c) 7860 tonnes.

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T08

08\_MODEL

08\_MODEL\_CORE

CHAPTER

**Modelling with Sequences & Series**

FAMILY

**Model setup**

VARIANT

**core family move****6 marks****WMA12/01 OCTOBER 2024****Question 7**

Past-paper question

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7. Jem pays money into a savings scheme,  $A$ , over a period of 300 months.

Jem pays £20 into scheme  $A$  in month 1, £20.50 in month 2, £21 in month 3 and so on, so that the amounts Jem pays each month form an arithmetic sequence.

- (a) Show that Jem pays £69.50 into scheme  $A$  in month 100 (1)

- (b) Find the **total** amount that Jem pays into scheme  $A$  over the period of 300 months. (2)

Kim pays money into a different savings scheme,  $B$ , over the same period of 300 months.

In a model, the amounts Kim pays into scheme  $B$  increase by the same percentage each month, so that the amounts Kim pays each month form a geometric sequence.

Given that Kim pays

- £20 into scheme  $B$  in month 1
- £250 into scheme  $B$  in month 300

- (c) use the model to calculate, to the nearest £10, the difference between the total amount paid into scheme  $A$  and the total amount paid into scheme  $B$  over the period of 300 months. (3)

T08 Modelling with  
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08\_MODEL Model setup

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family move

## WORKED SOLUTION

## Question 7

WMA12/01 October 2024

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1

**Identify Jem's sequence**

Jem's payments are arithmetic with first term 20 and common difference 0.50.

2

**Show month 100**The month 100 payment is  $20 + 99(0.50) = 69.50$ , so Jem pays £69.50.

3

**Sum Jem's payments**

$$S_{300} = \frac{300}{2} \{2(20) + 299(0.50)\} = 150(189.5) = 28425.$$

4

**Model Kim's payments**Let the common ratio be  $r$ . Since the first payment is 20 and the month 300 payment is 250,  $20r^{299} = 250$ .

5

**Find the ratio**Thus  $r^{299} = 12.5$  and  $r = 12.5^{1/299}$ .

6

**Sum Kim's payments**

$$S_B = 20 \frac{r^{300} - 1}{r - 1} \text{ with } r = 12.5^{1/299}, \text{ giving } S_B \approx 27362.95.$$

7

**Find the difference**The difference is  $28425 - 27362.95 \approx 1062.05$ , which is £1,060 to the nearest £10.**FINAL ANSWER**

(a) Proven. (b) £28,425. (c) £1,060.

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